Dittert's conjecture in dimension 8

An exact computer-assisted boundary-exclusion proof

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Status. This is an exact, reproducible working proof. It has not yet undergone independent peer review. The finite dimension-specific calculations are checked by two separately implemented standard-library verifiers.

Abstract

For a nonnegative $n \times n$ matrix A whose entries sum to n, let r_i, c_j be its row and column sums and put

$$\Phi(A) = \prod_i r_i + \prod_j c_j - \operatorname{per} A.$$

We prove Dittert's conjecture in dimension $n = 8$. A maximal Li scaling reduces the boundary problem to whole, marginal, and proper tight cuts. Most proper cuts are excluded by exact staircase-face bounds; the six one-dimensional-core cuts use sharper dimension-specific line-sum deficit estimates. For a marginal cut, stationarity forces a large column containing the minimum row. Cofactor nonnegativity bounds every other row meeting this column, while the zeroes in that column supply an additional staircase-face permanent bound. Six exact degree-65 Bernstein certificates exclude every possible nontrivial column support. No four-independent-zero theorem is used.

1 Statement and published inputs

Let

$$K_n = \left\{ A = (a_{ij}) \in \mathbb{R}^{n\times n}_{\geq 0} : \sum_{i,j} a_{ij} = n \right\}, \qquad \gamma_n = \frac{n!}{n^n}.$$

For $A \in K_n$, write

$$R = \prod_{i=1}^{n} r_i, \qquad C = \prod_{j=1}^{n} c_j, \qquad P = \operatorname{per} A.$$

Thus $\Phi(A) = R + C - P$ and $\Phi(J_n/n) = 2 - \gamma_n$.

Theorem 1.1. *For every $A \in K_8$,*

$$\Phi(A) \leq 2 - \frac{8!}{8^8},$$

with equality if and only if $A = J_8/8$.

We use the following published results.

1. Hwang proved that the only possible positive Φ-maximizer in K_n is J_n/n [5].
2. Cheon and Wanless proved that a partly decomposable matrix cannot maximize Φ, and that a maximizer cannot have its complete zero set equal, up to row and column permutations, to one proper rectangular block [1, Theorems 3.2 and 3.7].
3. Li's cut characterization says that a nonnegative $n \times n$ matrix S is doubly superstochastic if and only if
$$\sum_{i \in I, j \in J} s_{ij} \geq |I| + |J| - n$$
for all $I, J \subseteq [n]$ [6]; see also [1, Lemma 2.2].
4. On a staircase face of the Birkhoff polytope, the barycenter minimizes the permanent [4].
5. Minc's averaging theorem for permanent minimizers with repeated support rows and columns gives the block form used below [7, Theorem 1]; see also [8, Section 2].
6. The van der Waerden permanent theorem gives $\operatorname{per} B \geq \gamma_n$ for every doubly stochastic B, with equality only at J_n/n [2, 3].

The support-stationarity equation and the large-column support argument used below are proved directly; they are not additional external assumptions.

2 Boundary structure and the shared deficit

Because K_n is compact, Φ has a global maximizer. By Hwang's theorem, a nonuniform maximizer must lie on the boundary.

Lemma 2.1 (Two independent zeroes)**.** *Every boundary Φ-maximizer has two zero entries in distinct rows and distinct columns.*

Proof. Suppose a nonempty zero set contains no such pair. Fix a zero at (i, j). Every other zero shares row i or column j. If there were both a zero (i, j'), $j' \neq j$, and a zero (i', j), $i' \neq i$, those two zeroes would be independent. Hence all zeroes lie in one row or all lie in one column.

If that row or column is entirely zero, the matrix is partly decomposable. Otherwise its complete zero set is one proper rectangular block. Both cases are excluded for a maximizer by Cheon–Wanless. □

Assume for contradiction that a boundary maximizer A satisfies $\Phi(A) \geq 2 - \gamma_n$. Since the row sums and column sums each have arithmetic mean 1, AM–GM gives $R, C \leq 1$. Consequently $P \leq \gamma_n$. Write
$$P = \gamma_n - \delta, \qquad 0 \leq \delta \leq \gamma_n,$$
and put $R = 1 - \rho$, $C = 1 - \sigma$. The assumed lower bound gives
$$\rho + \sigma \leq \delta. \tag{1}$$
In particular,
$$R \geq 1 - \delta, \qquad C \geq 1 - \delta, \qquad RC \geq 1 - \delta. \tag{2}$$

3 The maximal Li scaling

For $I, J \subseteq [n]$, let $s(A[I,J])$ be the sum of the indicated submatrix, and define

$$\lambda = \min_{\substack{I,J\subseteq[n] \\ k:=|I|+|J|-n>0}} \frac{s(A[I,J])}{k}. \tag{3}$$

A maximizer is fully indecomposable by the Cheon–Wanless exclusion. Hence every numerator in (3) is positive, so $\lambda > 0$; taking $I = J = [n]$ shows $\lambda \le 1$.

Li's theorem says that A/λ is doubly superstochastic. Thus there is a doubly stochastic matrix B such that

$$\lambda B \le A \quad \text{entrywise}, \qquad P \ge \lambda^n \operatorname{per} B. \tag{4}$$

Choose I, J attaining (3), and set

$$u = n - |I|, \qquad v = n - |J|, \qquad k = n - u - v > 0.$$

There are three cases: the whole cut $(u,v) = (0,0)$, a marginal cut where exactly one of u, v is zero, and a proper cut where $u, v \ge 1$.

If $u = v = 0$, then $\lambda = 1$. Since $B \le A$ and both matrices have total sum n, $A = B$ is doubly stochastic. The van der Waerden theorem and $P \le \gamma_n$ force $A = J_n/n$, contrary to the boundary assumption.

4 The two-independent-zero permanent gap

Let $\Omega_n^{(2)}$ be the face of the Birkhoff polytope specified by $b_{12} = b_{21} = 0$, and put $m = n - 2$. Minc's averaging theorem permits a permanent minimizer on this face to be chosen in the form

$$Y = \begin{pmatrix} x_1 & 0 & a_1\mathbf{1}^\mathsf{T} \\ 0 & x_2 & a_2\mathbf{1}^\mathsf{T} \\ a_1\mathbf{1} & a_2\mathbf{1} & xJ_m \end{pmatrix}, \qquad x_i = 1 - ma_i, \qquad a_1 + a_2 + mx = 1. \tag{5}$$

Here $0 \le a_i \le 1/m$.

Proposition 4.1 (One-variable reduction)**.** *For $m \ge 4$, the minimum permanent in (5) occurs with $a_1 = a_2$. Consequently the minimum of* $\operatorname{per} B$ *on $\Omega_n^{(2)}$ is the minimum, on*

$$\frac{m-2}{m^2} \le x \le \frac{1}{m},$$

of

$$P_n(x) = m!x^{m-2}\left(x^2a^2 + 2mxab^2 + m(m-1)b^4\right), \tag{6}$$

where

$$a = \frac{2 - m + m^2x}{2}, \qquad b = \frac{1 - mx}{2}. \tag{7}$$

Proof. Put $s = a_1 + a_2$, $p = a_1a_2$, and $x = (1-s)/m$. Counting permanent terms in (5) gives, apart from the positive factor $m!x^{m-2}$,

$$E_m(s,p) = x^2(1 - ms + m^2p) + mx(s^2 - (2 + ms)p) + m(m-1)p^2.$$

For fixed s,

$$\frac{\partial E_m}{\partial p} = 2m(m-1)p + (m+1)s^2 - ms - 1.$$

Since $p \le s^2/4$ and $0 \le s \le 2/m$, this is at most

$$D_m(s) = \frac{(m^2+m+2)s^2 - 2ms - 2}{2}.$$

The quadratic is convex, while

$$D_m(0) = -1, \qquad D_m(2/m) = -1 + \frac{2}{m} + \frac{4}{m^2} < 0 \quad (m \ge 4).$$

Thus the derivative is strictly negative. For fixed s, the permanent is minimized by maximizing p, namely by $a_1 = a_2 = b$. Solving the stochastic constraints yields (7) and the stated interval. A second count gives (6). □

For $n = 8$, set

$$L_8 = \frac{12249}{5000000}. \tag{8}$$

Proposition 4.2 (Exact two-zero bound)**.** *Every* $B \in \Omega_8^{(2)}$ *satisfies*

$$\operatorname{per} B > L_8 > \gamma_8.$$

Proof. Here $m = 6$ and the complete interval is $[1/9, 1/6]$. The primary verifier expands $P_8(x) - L_8$ over $\mathbb{Q}$, verifies positive endpoint values, and uses an exact rational Sturm sequence to prove that it has no root on the interval.

The independent verifier instead evaluates the matrix (5) with (7) by the literal exact Ryser formula at nine rational points, interpolates the degree-eight polynomial, and proves positivity by exact Bernstein coefficients on 256 subintervals. Finally,

$$L_8 - \gamma_8 = \frac{476577}{10240000000} > 0. \tag{9}$$

□

5 Marginal tight cuts

We first record the stationarity equation used twice below.

Lemma 5.1 (Support stationarity)**.** *At a global maximizer, for every positive entry* a_{ij}*,*

$$\frac{R}{r_i} + \frac{C}{c_j} - \operatorname{per} A(i \mid j) = \Phi(A), \tag{10}$$

where $A(i \mid j)$ *is obtained by deleting row* i *and column* j*.*

Proof. The left side is $\partial\Phi/\partial a_{ij}$. Moving a small amount of mass between positive entries shows that all partial derivatives on the support have a common value μ. Euler's identity, using homogeneity of degree n, gives

$$n\Phi(A) = \sum_{i,j} a_{ij} \frac{\partial \Phi}{\partial a_{ij}} = \mu \sum_{i,j} a_{ij} = n\mu.$$

Thus $\mu = \Phi(A)$. □

Suppose the tight cut is marginal. Its ratio in (3) is an average of a nonempty set of row sums or column sums. Singleton marginal cuts are also allowed, so a minimizing marginal cut equals a minimum line sum. By transposition, choose a minimum row i with

$$r_i = \lambda = 1 - a, \qquad 0 \le a < 1.$$

Multiply (10) by a_{ij} and sum over j. Expansion of the permanent along row i gives

$$R + C \sum_j \frac{a_{ij}}{c_j} - P = r_i \Phi(A).$$

Therefore

$$q_i := \sum_j \frac{a_{ij}}{c_j} = 1 - (1+\theta)a, \qquad \theta = \frac{R-P}{C}. \tag{11}$$

By (2) and $P = \gamma_n - \delta$,

$$\theta \ge R - P \ge 1 - \gamma_n. \tag{12}$$

Let $c_{\max} = \max_j c_j$. Since $q_i > 0$,

$$\frac{1-a}{c_{\max}} \le q_i \le 1 - (2-\gamma_n)a.$$

Thus the last denominator is positive and

$$c_{\max} \ge 1 + h(a), \qquad h(a) = \frac{(1-\gamma_n)a}{1-(2-\gamma_n)a}. \tag{13}$$

AM–GM on the other seven row sums and column sums gives

$$R \le f(a) := (1-a)\left(1 + \frac{a}{7}\right)^7, \tag{14}$$

$$C \le g(h(a)) := (1+h(a))\left(1 - \frac{h(a)}{7}\right)^7. \tag{15}$$

Define

$$D(a) = 2 - f(a) - g(h(a)). \tag{16}$$

The shared deficit implies $\delta \ge D(a)$. The functions f and g are strictly decreasing on the relevant positive arguments, while h is increasing, so D is strictly increasing. The exact check

$$D(1/20) > \gamma_8 \tag{17}$$

confines every feasible value to $0 \le a < 1/20$.

The dominated doubly stochastic matrix B inherits the two independent zeroes of A, so Proposition 4.2 gives

$$P \ge L_8(1-a)^8. \tag{18}$$

The elementary scalar contradiction is

$$H(a) := L_8(1-a)^8 - \gamma_8 + D(a) > 0. \tag{19}$$

After clearing the positive denominator $7^7[1-(2-\gamma_8)a]^8$, its numerator has degree 16. The primary verifier uses exact Sturm sequences to prove $H > 0$ on

$$[0, 1/400] \ \cup \ [1/75, 1/20]. \tag{20}$$

The independent verifier reconstructs the numerator from exact values and proves the same statements by Bernstein positivity. It remains to treat

$$\frac{1}{400} \le a \le \frac{1}{75}. \tag{21}$$

5.1 The large-column support bound

The weighted-average identity

$$\frac{q_i}{r_i} = \sum_{j:a_{ij}>0} \frac{a_{ij}}{r_i}\frac{1}{c_j}$$

shows that some j_* with $a_{ij_*} > 0$ satisfies

$$c_{j_*} \geq \frac{r_i}{q_i} \geq 1 + h(a). \tag{22}$$

Thus the large column may be chosen to contain the minimum row.

Put

$$d(a) = \left(1 - \frac{h(a)}{7}\right)^7, \qquad q(a) = 2 - \gamma_8 - d(a). \tag{23}$$

For every ℓ with $a_{\ell j_*} > 0$, cofactor nonnegativity and Lemma 5.1 give

$$\frac{R}{r_\ell} = \Phi(A) - \frac{C}{c_{j_*}} + \operatorname{per} A(\ell \mid j_*) \geq 2 - \gamma_8 - d(a) = q(a).$$

Indeed, C/c_{j_*} is the product of the other seven column sums, whose sum is at most $7 - h(a)$. On (21), q is increasing and the verifier checks exactly that $q(1/400) > 1$. Since $R \leq 1$,

$$r_\ell \leq \frac{R}{q(a)} \leq \frac{1}{q(a)} < 1 \qquad (a_{\ell j_*} > 0). \tag{24}$$

Let z be the number of zero entries in column j_*. Full indecomposability excludes a column with only one positive entry, while (24) excludes a full-support column. Hence $1 \leq z \leq 6$. The support consists of the distinguished row of sum $1 - a$ and $7 - z$ other rows bounded by $1/q(a)$. Optimizing the remaining z row sums by AM–GM gives

$$R \leq U_z(a) := (1-a)q(a)^{-(7-z)}\left(\frac{7 + a - (7-z)/q(a)}{z}\right)^z. \tag{25}$$

The objective is increasing in each capped row sum: after the uncapped rows are equalized, their mean is greater than 1, whereas every capped row is less than 1.

The matrix B also inherits the $z \times 1$ zero rectangle in column j_*. Hwang's staircase theorem therefore gives

$$\operatorname{per} B \geq M_z := \max\left\{L_8, \frac{\gamma_{8-z}\gamma_7}{\gamma_{7-z}}\right\}. \tag{26}$$

The six exact values used by the verifier are

$$\left(\frac{12249}{5000000}, \frac{15625}{6353046}, \frac{36864}{14706125}, \frac{1215}{470596}, \frac{320}{117649}, \frac{360}{117649}\right).$$

Consequently

$$R \geq R_{0,z}(a) := 2 - \gamma_8 + M_z(1-a)^8 - g(h(a)). \tag{27}$$

Here is the exact polynomial certificate for $R_{0,z} > U_z$. Put

$$\begin{aligned}
e &= 1 - (2-\gamma_8)a, \qquad\qquad t = e - \frac{(1-\gamma_8)a}{7}, \\
Q &= (2-\gamma_8)e^7 - t^7 = e^7 q(a), \\
S_z &= (2 - \gamma_8 + M_z(1-a)^8)e^8 - (e + (1-\gamma_8)a)t^7 = e^8 R_{0,z}(a), \\
W_z &= (7+a)Q - (7-z)e^7.
\end{aligned}$$

All denominators are positive on (21), and the desired inequality is equivalent to

$$G_z(a) := z^z Q^7 S_z - (1-a)e^{8+7(7-z)} W_z^z > 0. \tag{28}$$

Each G_z has degree 65. The primary verifier expands it directly and converts it to the degree-65 Bernstein basis on $[1/400, 1/75]$; every one of the $6 \cdot 66 = 396$ coefficients is strictly positive. The smallest is approximately $7.1630735 \cdot 10^{-6}$. The independent audit reconstructs each G_z from 66 exact values and obtains the same positivity conclusion. This contradicts (25)–(27) for every possible support and excludes all marginal cuts.

6 Proper tight cuts

Suppose now that $u, v \geq 1$. Define

$$\varepsilon_r = \sum_{i \in I^c} r_i - u, \qquad \varepsilon_c = \sum_{j \in J^c} c_j - v.$$

For positive numbers $x_1, \ldots, x_n$ of sum n and product X, binary Pinsker after grouping a subset and its complement gives

$$\left(\sum_{i \in S} x_i - |S| \right)^2 \leq -\frac{n}{2} \log X.$$

Applying this to the row and column sums, then using Cauchy–Schwarz and (2), yields

$$|\varepsilon_r| + |\varepsilon_c| \leq \sqrt{-n \log(RC)} \leq \sqrt{\frac{n\delta}{1-\delta}} =: T. \tag{29}$$

The cut identity

$$s(A[I,J]) = k - \varepsilon_r - \varepsilon_c + s(A[I^c, J^c])$$

therefore implies

$$\lambda \geq 1 - \frac{T}{k}. \tag{30}$$

For $n = 8$, the exact inequality $n\gamma_n < 1 - \gamma_n$ ensures $T < 1 \leq k$.

Because B is doubly stochastic,

$$s(B[I,J]) = k + s(B[I^c, J^c]) \geq k.$$

Tightness and (4) give the reverse bound after multiplication by λ, so equality holds and $B[I^c, J^c] = 0$. Thus B lies on the staircase face with a prescribed $u \times v$ zero block. Hwang's theorem gives

$$\operatorname{per} B \geq \nu_{n;u,v} := \frac{\gamma_{n-u}\gamma_{n-v}}{\gamma_k}, \qquad k = n - u - v, \tag{31}$$

where $\gamma_0 = 1$. The barycenter has $(n-v)!(n-u)!/k!$ supported permutations, all of the same weight; both verifiers reconstruct this count exactly.

Using (30), Bernoulli's inequality, and $1 - \delta \geq 1 - \gamma_n$ gives

$$\nu_{n;u,v} \left(1 - \frac{T}{k}\right)^n - (\gamma_n - \delta) \geq \nu_{n;u,v} - \gamma_n + \delta - \frac{n\nu_{n;u,v}}{k} \sqrt{\frac{n\delta}{1-\gamma_n}}.$$

Completing the square in $\sqrt{\delta}$ gives the uniform lower bound

$$\eta_{n;u,v} := \nu_{n;u,v} - \gamma_n - \frac{n^3 \nu_{n;u,v}^2}{4k^2(1-\gamma_n)}. \tag{32}$$

Both verifiers check $\eta_{8;u,v} > 0$ for 15 of the 21 proper cuts

$$u, v \geq 1, \qquad u + v \leq 7.$$

The smallest positive value occurs at $(u, v, k) = (1, 1, 6)$, where

$$\nu_{8;1,1} = \frac{33592320}{13841287201}, \qquad \eta_{8;1,1} = \frac{890634308697114685O328025}{328343054865251993489657783910 4} > 0.$$

The only cuts not settled by (32) have $k = 1$, namely $(u, v) = (1, 6), (2, 5), (3, 4)$ and their transposes. We treat them with a sharper product estimate.

For $1 \leq d \leq 7$, define

$$F_d(x) = \left(1 + \frac{x}{d}\right)^d \left(1 - \frac{x}{8-d}\right)^{8-d}. \tag{33}$$

If a d-element subset of the row sums has sum $d + x$, AM–GM separately within the subset and its complement gives $R \leq F_d(x)$; the analogous statement holds for C. Put

$$(c_1, c_2, c_3, c_4, c_5, c_6, c_7) = \left(\frac{1}{2}, \frac{3}{10}, \frac{1}{4}, \frac{6}{25}, \frac{1}{4}, \frac{3}{10}, \frac{1}{2}\right). \tag{34}$$

The exact polynomial inequalities

$$1 - F_d(x) \geq c_d x^2 \qquad \left(-\frac{1}{10} \leq x \leq \frac{1}{10}, \quad 1 \leq d \leq 7\right) \tag{35}$$

are checked as follows. After removing the double zero at $x = 0$, each $1 - F_d(x) - c_d x^2$ has a degree-six quotient. Both verifiers convert that quotient to the Bernstein basis on $[-1/10, 1/10]$ and find every coefficient strictly positive; the smallest coefficient is

$$\frac{640006399}{65536000000} > 0.$$

They also check $1 - F_d(\pm 1/10) > \gamma_8$ for every d. Since F_d is strictly increasing to 1 on the negative side of zero and strictly decreasing from 1 on the positive side, the lower bounds $R, C \geq 1 - \delta \geq 1 - \gamma_8$ localize every relevant subset deviation to $(-1/10, 1/10)$.

Now take one of the six remaining cuts and write

$$x = \varepsilon_r, \qquad y = \varepsilon_c, \qquad s = x + y, \qquad u + v = 7.$$

The cut identity gives $\lambda \geq 1 - s$. If $s \leq 0$, then $\lambda \geq 1$; as $\lambda \leq 1$, the domination argument makes A doubly stochastic and contradicts the boundary assumption. Hence $s > 0$. From (1), (33), and (35),

$$\delta \geq \rho + \sigma \geq c_u x^2 + c_v y^2 \geq \frac{c_u c_v}{c_u + c_v} s^2 =: c_{u,v} s^2. \tag{36}$$

For the three cases up to transposition,

$$c_{1,6} = \frac{3}{16}, \qquad c_{2,5} = \frac{3}{22}, \qquad c_{3,4} = \frac{6}{49}. \tag{37}$$

The weakest of these constants and the exact inequality $2401\gamma_8 < 6$ imply $s < 1/7$.

Here the staircase core has size one, so (31) becomes $\nu_{8;u,v} = \gamma_{8-u}\gamma_{8-v}$. Therefore

$$P \geq \nu_{8;u,v}(1-s)^8, \qquad P = \gamma_8 - \delta \leq \gamma_8 - c_{u,v}s^2.$$

For $(u,v) = (1,6), (2,5), (3,4)$, both verifiers prove

$$K_{u,v}(s) := \nu_{8;u,v}(1-s)^8 - \gamma_8 + c_{u,v}s^2 > 0 \qquad \left(0 \leq s \leq \frac{1}{7}\right) \tag{38}$$

by exact degree-eight Bernstein coefficients on the two intervals $[0, 1/14]$ and $[1/14, 1/7]$. Transposition handles the other three cuts. The smallest coefficient among these certificates is

$$\frac{13183225395}{259172170858496} > 0.$$

Thus (38) contradicts the preceding upper bound for P, and all proper cuts are excluded.

The whole, marginal, and proper cases exhaust every tight cut. No boundary global maximizer exists. Hwang's positive-support theorem identifies the sole remaining maximizer as $J_8/8$, proving Theorem 1.1.

7 Exact computational audit

No floating-point value is used in a correctness decision. The primary standard-library verifier checks:

1. the bivariate two-zero equalization identity and its sign;
2. the degree-eight permanent polynomial and the lower bound (8) by a rational Sturm sequence;
3. denominator and endpoint signs, (17), and zero Sturm roots for the degree-16 numerator on both safe intervals;
4. $q(1/400) > 1$, all six support bounds M_z, and all 396 exact Bernstein coefficients in (28);
5. all seven subset-deficit inequalities (35), their endpoint localization checks, and the three polynomials (38);
6. the staircase barycenter count for all 21 proper cuts and all 15 positive values in (32).

The independent verifier imports no code from the primary verifier. It uses a literal exact Ryser sum for the two-zero matrices and Bernstein positivity on 256 subintervals. It reconstructs the marginal numerator from 17 exact values and each support polynomial from 66 exact scalar values. It also reconstructs each subset quotient from seven values and each exceptional-cut polynomial from nine values, checks off-grid values, and proves positivity by an independently implemented Bernstein conversion. Thus none of the principal polynomial expansions is shared.

The integration test runs both programs in ordinary and optimized Python mode and requires identical output. Three mutations inflate L_8, omit the staircase support improvement in the marginal argument, and replace $c_4 = 6/25$ by the false value $1/4$; the appropriate independently mutated program must reject each change.

8 Scope and status

This note establishes dimension 8. Together with the other exact working proofs in the accompanying repository, the covered dimensions are

$$4, 5, 8, 9, 10, 11, 12, 13, 14, 15, 16.$$

Dimensions $6, 7$ are not settled by these materials. The present result should be described as an exact computer-assisted working proof until the mathematical reduction and both programs have undergone independent peer review.

Reproducibility

From the `n8` directory, run

```
python3 -I verify_dittert_n8.py
python3 -I audit_dittert_n8_stdlib.py
python3 -I test_n8_package.py
```

The verifier final lines must be

```
CERTIFIED
INDEPENDENT AUDIT CERTIFIED
```

and the tests must report `OK`.

References

[1] G.-S. Cheon and I. M. Wanless, *Some results towards the Dittert conjecture on permanents*, Linear Algebra Appl. 436 (2012), 791–801. doi:10.1016/j.laa.2010.08.041.

[2] G. P. Egorychev, *Solution of the van der Waerden problem for permanents*, Soviet Math. Dokl. 23 (1981), 619–622.

[3] D. I. Falikman, *A proof of van der Waerden's conjecture on the permanent of a stochastic matrix*, Mat. Zametki 29 (1981), 931–938.

[4] S.-G. Hwang, *Minimum permanent on faces of staircase type of the polytope of doubly stochastic matrices*, Linear Multilinear Algebra 18 (1985), 271–306. doi:10.1080/03081088508817694.

[5] S.-G. Hwang, *A note on a conjecture on permanents*, Linear Algebra Appl. 76 (1986), 31–44. doi:10.1016/0024-3795(86)90212-0.

[6] C.-K. Li, *On certain convex matrix sets*, Discrete Math. 79 (1990), 323–326.

[7] H. Minc, *Minimum permanents of doubly stochastic matrices with prescribed zero entries*, Linear Multilinear Algebra 15 (1984), 225–243. doi:10.1080/03081088408817592.

[8] K. Pula, S.-Z. Song, and I. M. Wanless, *Minimum permanents on two faces of the polytope of doubly stochastic matrices*, Linear Algebra Appl. 434 (2011), 232–238. doi:10.1016/j.laa.2010.08.022.